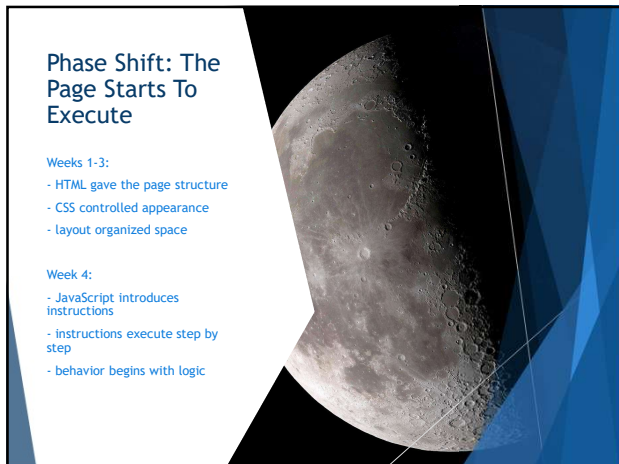
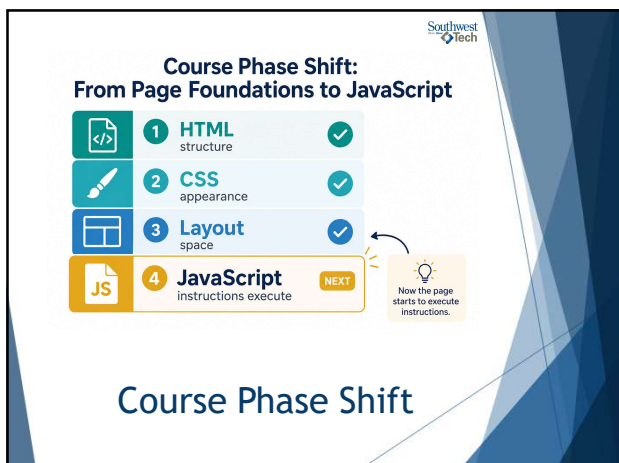




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3

Lab Review: Successful Path

One successful path:

- ▶ Flexbox used intentionally
- ▶ spacing and alignment improved
- ▶ at least one media query
- ▶ layout adapts on smaller screens
- ▶ navigation remains usable

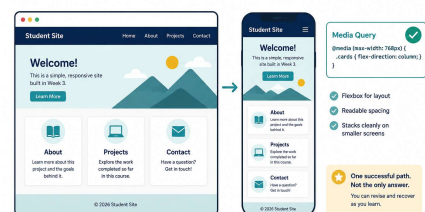
This completes the first foundation layer.



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Week 3: One Successful Responsive Layout Path

Built with Flexbox • Readable spacing • Clean content sections



Review: Assignment 3

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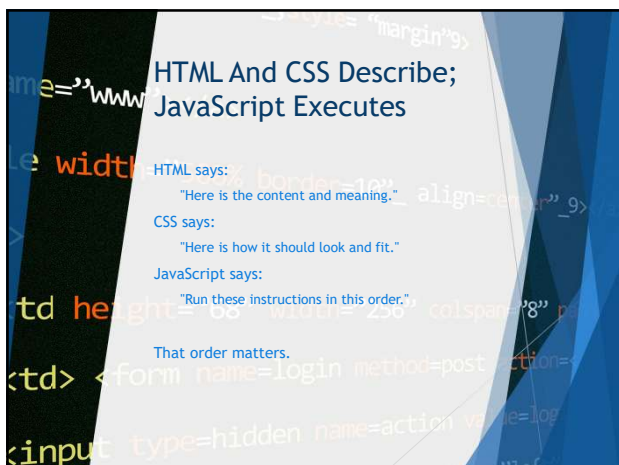
HTML And CSS Describe; JavaScript Executes

HTML says:
"Here is the content and meaning."

CSS says:
"Here is how it should look and fit."

JavaScript says:
"Run these instructions in this order."

That order matters.



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HTML, CSS, and JavaScript: What They Do

Three parts. Different jobs. They work together to build the web.

HTML

Describes structure. Defines what content exists and how it is organized.

```
<html>
  <head>
    <title>Site /title>
  </head>
  <body>
    <p>Welcome!</p>
    <ul>
      <li>About</li>
      <li>Projects</li>
      <li>Contact</li>
    </ul>
  </body>
</html>
<footer>© 2020</footer>
```

CSS

Describes appearance. Controls colors, fonts, spacing, and layout.

```
body {
  font-family: system-ui;
  color: #1f78b4;
}

header {
  background: #e6e6fa;
  color: white;
  padding: 10px;
}

main ul {
  display: flex;
  gap: 10px;
}
```

JavaScript

Executes instructions. Runs step-by-step logic to perform tasks.

Example Instruction Path

- 1 Define starting values.
- 2 Check a condition.
- 3 Do something based on the result.
- 4 End when complete.

Order matters when instructions run. JavaScript follows the order you write.

Describe vs. Execute

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What Counts as Success Today

- a `.js` file runs
- output appears in the console
- variables hold values
- an `if / else` chooses a path
- you predict before you run

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What We Will Use Today

- `script.js`
- script link
- variable
- value
- Boolean
- `console.log()`
- `if / else`
- browser console

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Today's Beginner JavaScript Toolbox

Simple tools. Clear purpose. Strong foundation.

Use these tools to write, run, and see results.

Small steps. Clear results. You've got this!

Practice: write a little, run it, and check the console. Build confidence one step at a time.

Today's Toolbox

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What We Will "Skip" For Now

- ▶ finding page elements
- ▶ button clicks
- ▶ changing visible page text
- ▶ loops
- ▶ jQuery
- ▶ full app behavior

Today: run small logic and read the result.

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Topics We'll Return To Later

Important tools. Just not today.

Not today. First, run small logic and read the result.

Parked For Later

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What A Program Does

A program is a set of instructions.
The browser runs those instructions:

from top to bottom using stored values making decisions when conditions appear producing output we can inspect

Programming begins with mental execution.

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A Program Is Step-by-Step Instructions

The computer does exactly what you tell it, in the order you tell it.

1 Store a value
Save information to use later.

2 Print a value
Show information as output.

3 Check a condition
Ask a yes/no question.

4 Choose a path
Do one thing or another based on the answer.

Programming begins with mental execution.

Program As Instructions

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Variables Store Values

A variable gives a name to a value.

Example:

```
1 const assignmentScore = 84;
2 const submittedOnTime = true;
3 |
```

Read it as:
"remember this value using this name."

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Console Output Makes Thinking Visible

The console is our first output space.

`console.log()` helps us see:

- ▶ what value is stored
- ▶ which line ran
- ▶ what decision happened
- ▶ whether our prediction was right

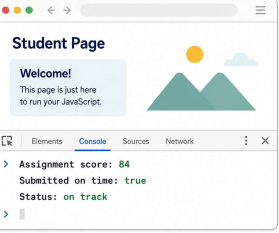
Today, console output is the visible result.

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The Browser Console: Today's Output Space

Use the console to see the results of your code.



Console Output

Console output makes thinking visible.

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Conditions Choose A Path

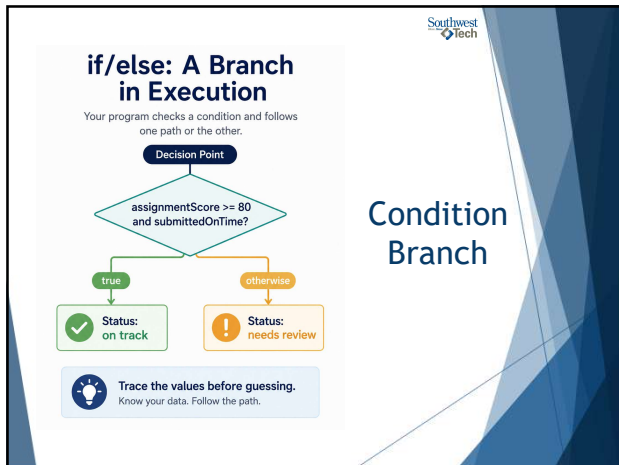
A condition creates a branch.

`'if'` means:
"when this is true, run this block."

`'else'` means:
"otherwise, run this other block."

Do not guess. -- Trace the values.

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What to Watch For

- ▶ `'script.js'` linked to the page
- ▶ variables created first
- ▶ `'console.log()'` output
- ▶ one `'if / else'` decision
- ▶ changing a value and predicting the result

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Live Demo: Values And Conditions

We'll write simple code. Then we'll read the result.

Code Workspace (script.js)

```
1 // 1. Variables
2 let assignmentScore = 94;
3 let submittedOnTime = true;
4
5 // 2. Condition
6 if (assignmentScore >= 80 && submittedOnTime) {
7   console.log("Status: on track");
8 } else {
9   console.log("Status: needs review");
10 }
11
```

Browser Console

```
> Status: on track
```

Change one value. Predict the result.

Demo: Values And Conditions

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Predict, Run, Observe

Use this rhythm:

1. predict what should happen
2. run or refresh the page
3. observe the console output
4. compare prediction to result
5. change one value and repeat

This is how logic becomes less mysterious.

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Week 4 JavaScript Learning Rhythm

A simple cycle that builds understanding.

1 predict
What do you think will happen?

2 run
Run your code.

3 observe
Read the output. What happened?

4 compare
Compare the result to your prediction.

5 change one value
Change one value. Keep everything else the same.

Repeat. Learn. Adjust. Improve.

Logic becomes clearer when you slow down.

Predict - Run - Observe

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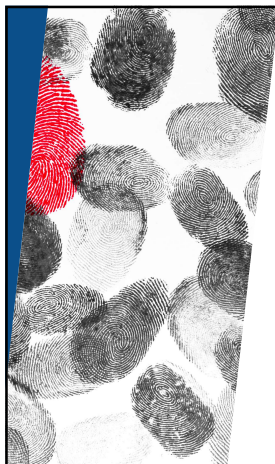
Lab

First JavaScript program
Create a `.js` file with:

- ▶ - at least 2 variables
- ▶ - at least 1 `if / else`
- ▶ - console output
- ▶ - a simple calculation or decision

*Functions are coming next;
start with values and
decisions.*

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Evidence

Preserve evidence:

- ▶ your `.js` file runs
- ▶ console output is visible
- ▶ variable names are meaningful
- ▶ the condition makes sense
- ▶ you can explain what changed when a value changed

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Closing Success Check

Today:

- ▶ JavaScript ran as instructions
- ▶ variables stored values
- ▶ console output made logic visible
- ▶ a condition chose a path

Next:

- ▶ we organize logic into functions.

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